

CSE5026: Introduction and Frontiers of Cognitive Science

Memory (I)

Yang Xu 徐炆

Department of Computer Science and Engineering
Southern University of Science and Technology

Overview

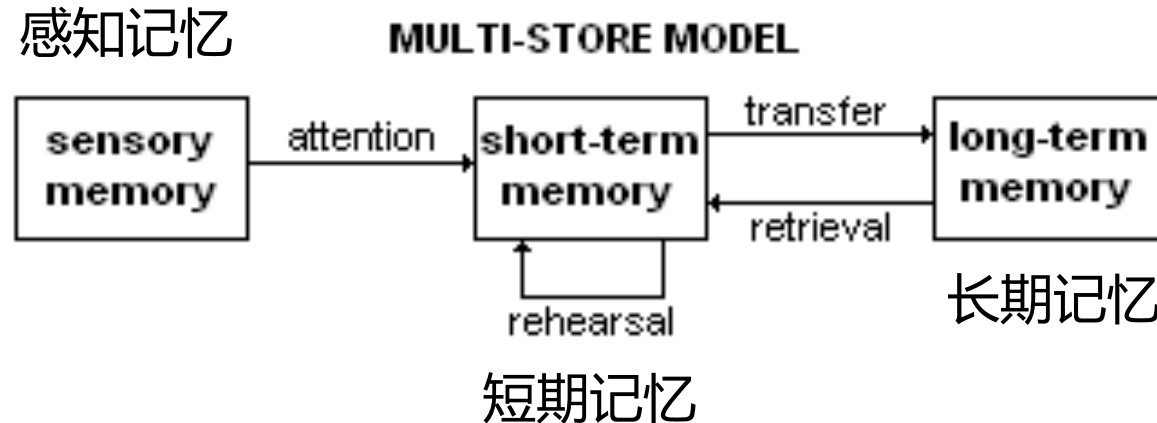
- Memory System 记忆系统
- Short-Term Memory 短时记忆
- Working Memory 工作记忆
- Memory Mechanism in AI/LLMs 大语言模型中的记忆机制

Introduction and Brief History

- 记忆是在人脑中积累、保存和提取个体经验的心理过程。
- 运用信息加工的术语，记忆就是人脑对外界输入的信息进行**编码、存储、和提取**的过程。
- 记忆是心理在时间上的持续，有了记忆，先后的经验才能联系起来，使心理活动成为一个发展的、统一的过程。

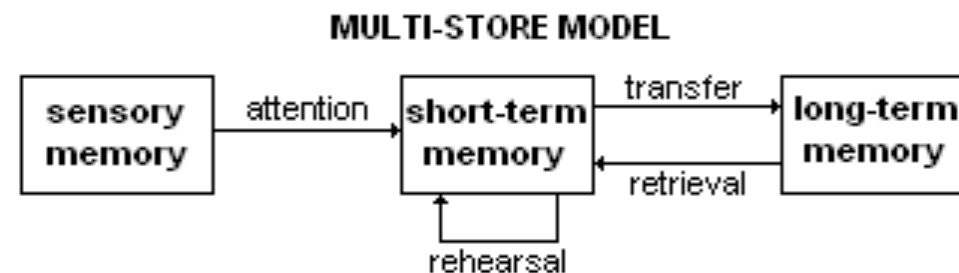
记忆系统的模型

- 常识：人类的记忆过程分为三个阶段：信息编码 (习得), 信息存储 (保持), 信息提取。
- Atkinson-Shiffrin's multi-store model of memory (1968)
- 多存储 (器) 模型。
- A simple linear model of human memory



Source: <https://en.wikipedia.org/wiki/Memory>

记忆的多存储器模型

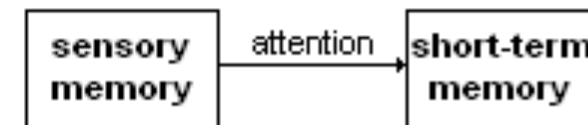


- A&S's 多存储器模型
- 基本假设：人的记忆系统中存在三种不同类型的**记忆存储器** (抽象，非实体器官)
- 感觉寄存器 (Sensory Register, SR)
- 短时记忆存储 (Short-Term Storage, STS)
- 长时记忆存储 (Long-Term Storage, LTS)

Source: https://en.wikipedia.org/wiki/Atkinson-Shiffrin_memory_model - Sensory_register

多存储器模型 - 感觉寄存器 SR

- 当感官感知到环境刺激时，感知信息会短暂地存储在 A&S 模型中的 **感觉寄存器 (SR)** 中，也称**感觉缓冲区** (sensory buffer)，或感觉记忆 (sensory memory)，或瞬时/即时记忆。
- 每种感官对应一个寄存器 (大多数研究集中在视觉和听觉)。
- 寄存器可以防止海量信息淹没高级认知功能。
- 只有当注意力集中时，信息才会转移到短期记忆。
- 否则信息会迅速衰减并被遗忘。



Source: https://en.wikipedia.org/wiki/Atkinson-Shiffrin_memory_model - Sensory_register

视觉寄存器 => 映像记忆 Iconic Memory

- 映像记忆 (iconic memory) 与视觉系统相关联。
 - 直观例子：“视觉残留”——当你快速移动点燃的焰火时，会看到一条“连续的光带”，而非单个的光点，说明感知到的视觉信息没有马上消失。



映像记忆 Iconic Memory 实验

- Sperling (1960) 部分报告法实验。
- 向被试快速呈现一张 “3 行 × 3 列” 的字母卡片（共 9 个字母），呈现时间仅 0.05 秒（远短于清晰识别字母所需时间）。
- 若要求被试 “报告所有看到的字母”，多数人仅能回忆 4-5 个。

R	N	P
A	B	D
E	Y	Q

shown for <50 ms

Source: <https://doi.org/10.1037%2Fh0093759>

映像记忆 Iconic Memory 实验

- Sperling (1960) 部分报告法实验。
- 若改为 “部分报告法”：呈现卡片后，立即用 “高音、中音、低音” 三种声音信号提示被试，分别报告 第1、2、3行的 字母。
- 结果显示，被试几乎能 100% 正确报告任意一行的 3 个字母。
- 说明**9 个字母的完整视觉信息曾短暂保存在映像记忆中**，只是未被**及时注意**的部分迅速消失。

R	N	P
A	B	D
E	Y	Q

shown for <50 ms

Source: <https://doi.org/10.1037%2Fh0093759>

听觉寄存器 => 声像记忆 Echoic Memory

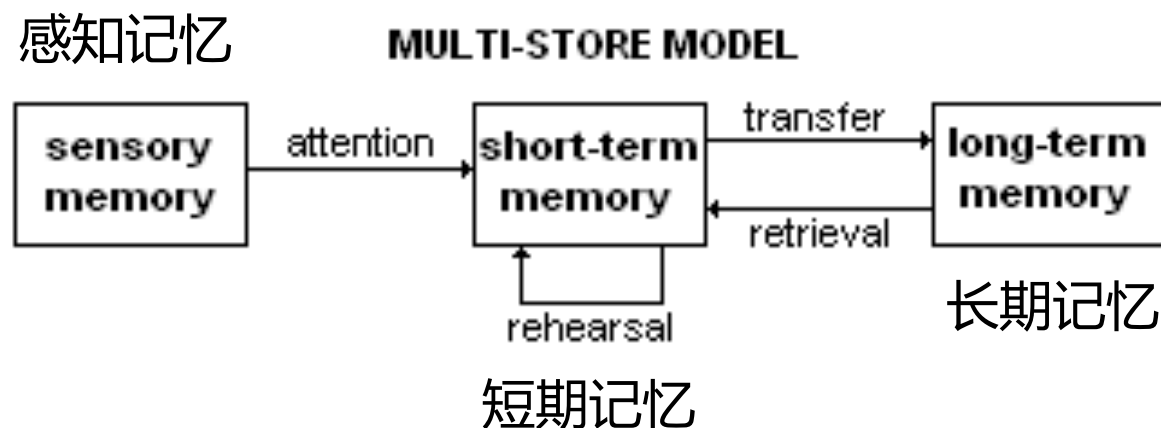
- 指留在听觉系统中的信息得以保留一段时间。
 - 例：一首乐曲结束后，仍然能在脑海中短暂地回响起刚才乐曲的旋律、节奏和器乐（余音绕梁、余韵）。
- 长度 1.5~5s，在没有干扰信息时可达 20s。

Source: https://en.wikipedia.org/wiki/Atkinson-Shiffrin_memory_model - Echoic_memory

记忆的层次

- 对刺激的加工与保持方面：
 - 映像记忆保持 1s 左右；声像记忆保持 2~4s。
 - 短时记忆：几秒到几十秒；长时记忆：几天、几个月、几年甚至终生。
- 信息存储的容量方面：
 - 短时记忆的信息加工容量平均为7个组块。
 - 长时记忆存储器的信息容量基本是无限的。

短时记忆



- 短时记忆是感知记忆和长期记忆的中间阶段，保持时间大约 5s 至1m。
- 信息保持时间短；记忆容量有限，一般为 7 ± 2 ；
- 短时记忆的信息可以被**主观意识**报告；通过复述可转入长时记忆。

神奇的数字: 7 ± 2

- George Miller (1956) 对短期记忆能力进行了定量研究。
- 年轻人的记忆广度大约为 7 个单位 (数字、字母、单词或其它), 称为组块 (chunk)。
- 后来的研究发现, 记忆容量与组块的类别有关:
 - 数字为 7 个, 字母为 6 个, 单词为 5 个; 较长词汇的记忆容量低于较短词汇。
- 后续研究 Cowan (2001) 认为年轻人的记忆能力为 4 个组块。

神奇的数字: 7 or 4?

- Cowan (2001) 主张短期记忆的容量要低得多, 挑战了 “ 7 ± 2 ” 的旧观念。
- 认为短时记忆容量所考察的更多是 “注意力集中” (focus of attention) 的能力, 仅在特定条件下会显现 (如当组块或复述等其他记忆过程被阻断)。
 - 主动的注意不同于被动的存储。
- 将信息分块 (如将电话号码分成几组数字) 可以扩展短期记忆, 但保存 “块” (chunk) 的底层容量仍然大约为 4 个。

工作记忆 Working Memory

- Baddeley & Hitch 在模拟短时记忆障碍的实验基础上提出了**工作记忆 (working memory)** 的概念。
- 工作记忆是一种对信息进行暂时处理和储存的系统，容量有限。在许多复杂的认知活动中起重要作用。
- 是对 A&S's 多存储模型的改进和增强。

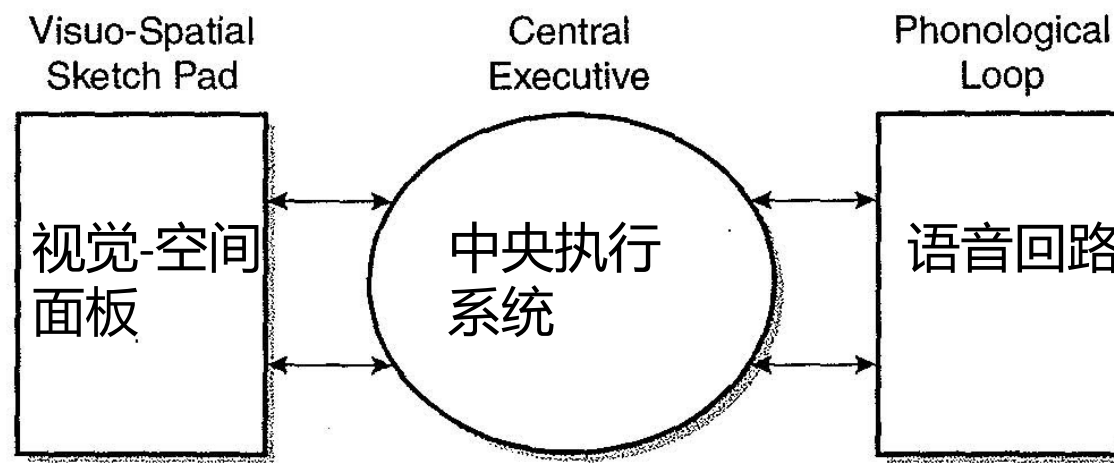
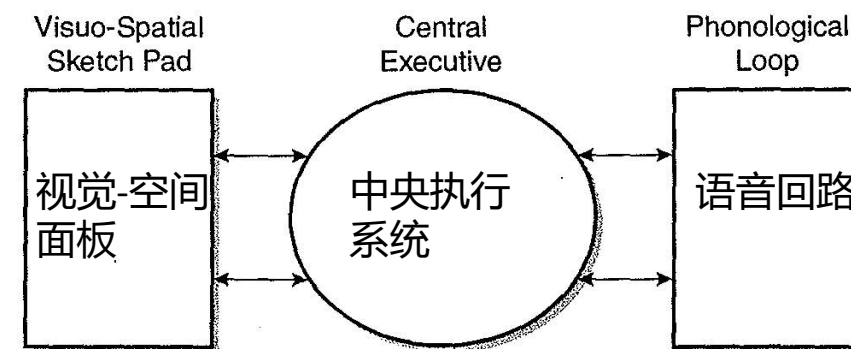


Figure from Baddeley (1986)

Source: https://en.wikipedia.org/wiki/Baddeley%27s_model_of_working_memory

工作记忆模型的主要组件



- **中央执行** (central executive): 作为监督系统，控制从属系统之间的信息流动。
- **语音回路**: 在整体上处理声音或语音信息，存储语言内容。
- **视觉-空间面板**: 处理视觉-空间信息，可以与语音回路同时处理听觉和视觉刺激，而不会相互影响。
- 2000年改进版模型加入了 **情景缓冲区 episodic buffer** 组件：主要功能是将不同领域的信息关联起来，形成具有时间顺序的视觉、空间和语言信息整合单元。(例如对故事或电影场景的记忆)

为什么提出工作记忆模型

- 基于对 “双任务范式” (dual-task) 实验结果的总结
 - 人在同时处理两个独立感知域 (视觉任务和语言任务) 的任务时, 其效率几乎与单独执行这些任务一样高。
 - 相反, 当人试图同时执行同一感知域下的另一个任务时, 其效率低于单独执行这些任务。
 - E.g. 边骑马边背诗 (易), 边背诗边写作文 (难)
- 中央执行的必要性: 阿尔茨海默病患者的中央执行功能正在衰退, 表现为: 单个任务完成地好, 但执行两个或多个动作变得困难。

Source: https://en.wikipedia.org/wiki/Baddeley%27s_model_of_working_memory

Overview

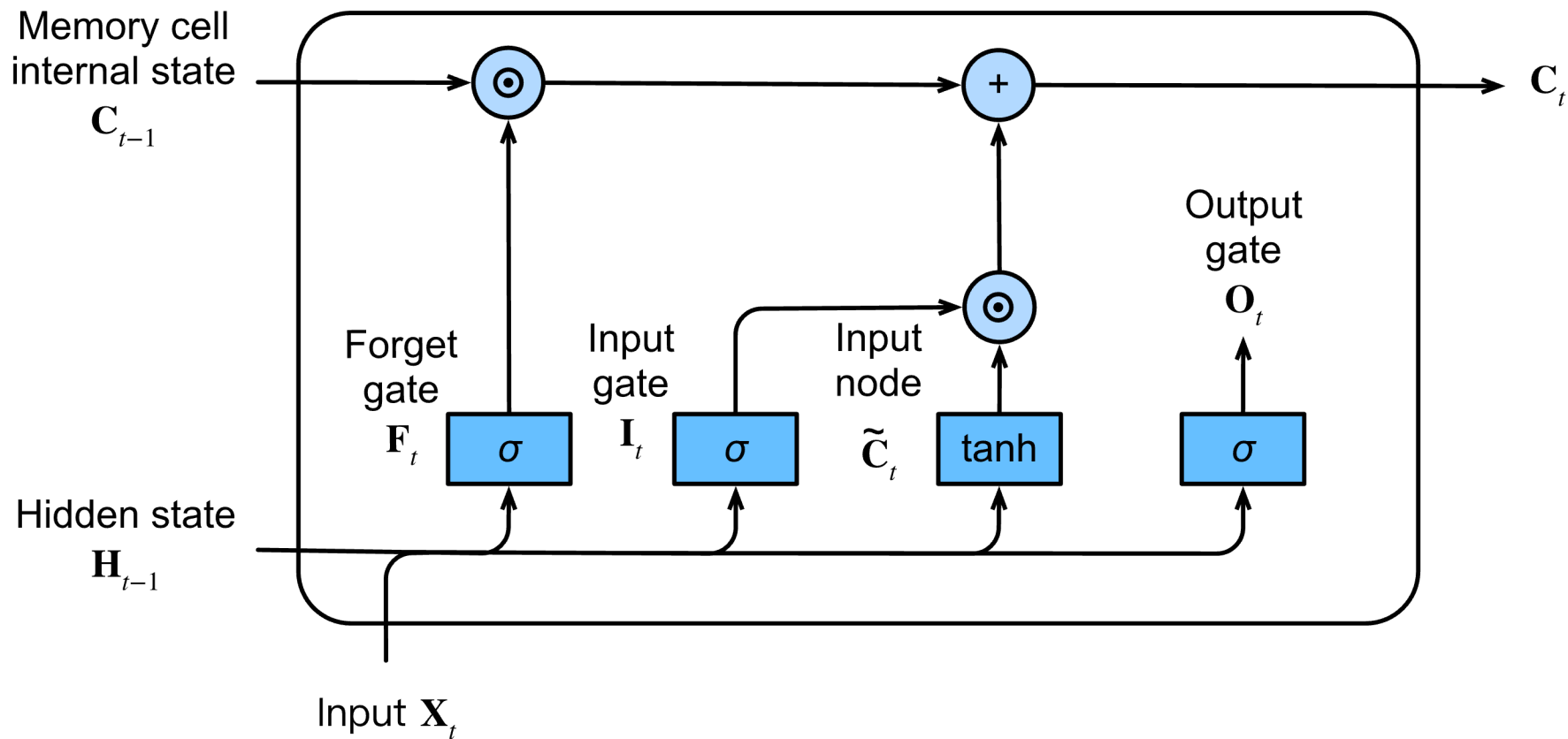
- Memory System 记忆系统
- Short-Term Memory 短时记忆
- Working Memory 工作记忆
- **Memory Mechanism in AI/LLMs 大语言模型中的记忆机制**

人工智能与记忆

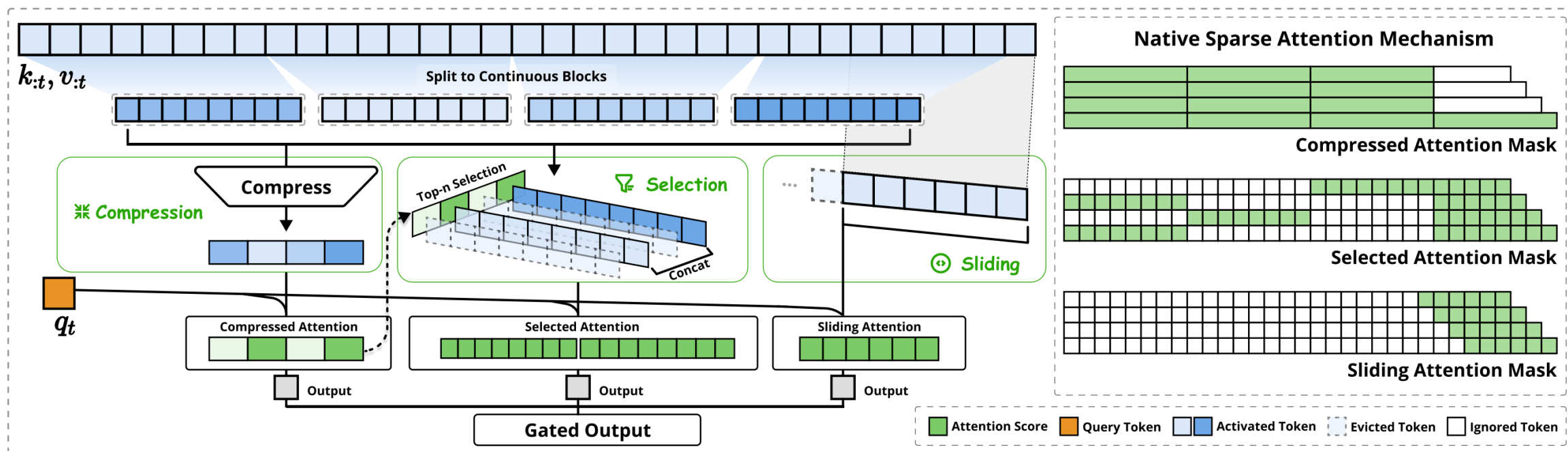
- 智能 or 记忆?
- 人工智能发展早期，追求可拓展性，努力排斥“记忆”在完成任务中发挥的作用。
 - E.g., train-val-test 避免过拟合。
- 近期，又开始追求特异性，努力构建 Agent 使用“记忆”来配合人类特殊需求。
 - E.g., 保留长上下文中的特定信息。

Simple “memory” mechanism in LM

- Long Short-Term Memory (LSTM)



Attention and Sparse Attention

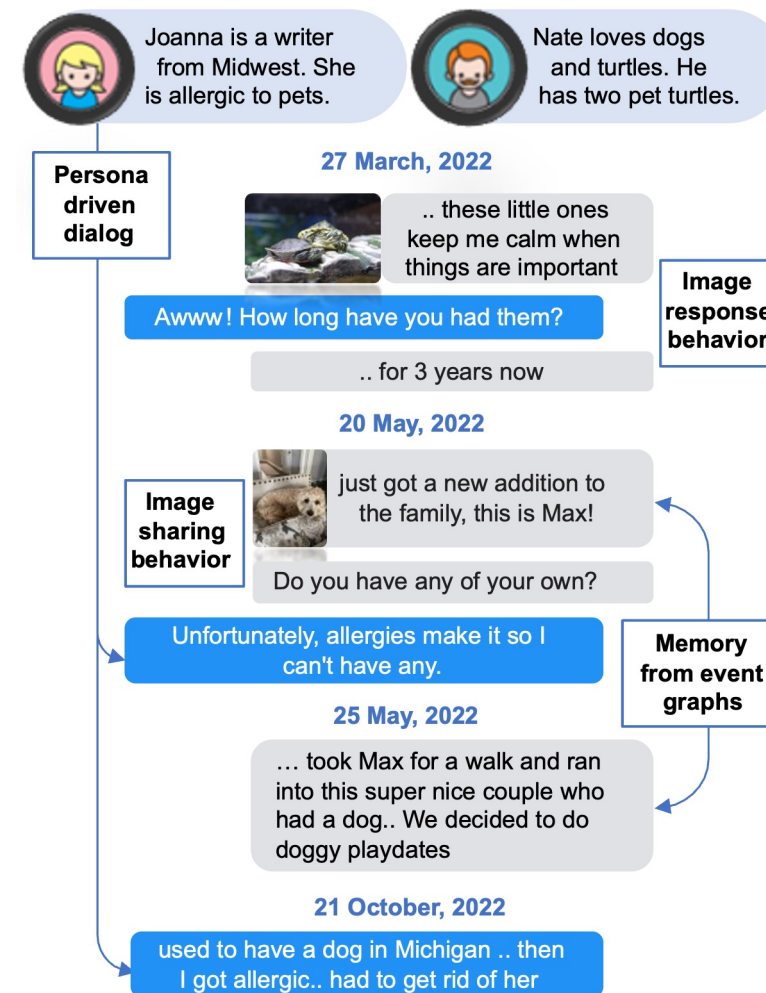


q_t : t 位置 token 的查询向量, $k_{:t}, v_{:t}$: 之前位置 token 的 KV 键值数据

Long-Term Conversation

Source: <https://snap-research.github.io/locomo/>

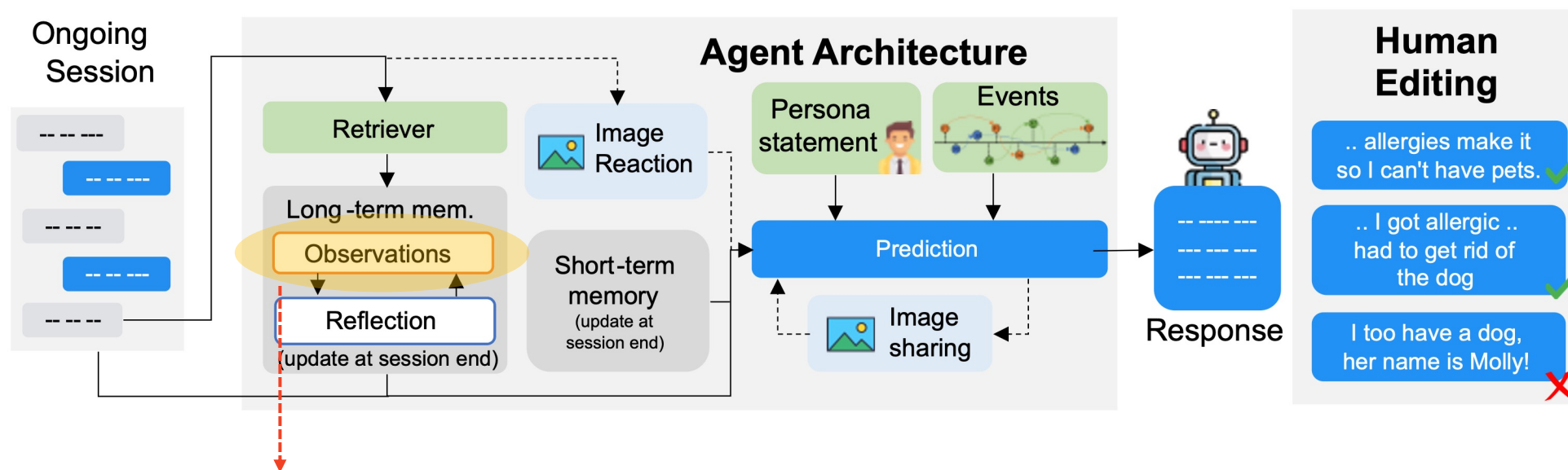
- **LoCoMo**: A dataset of very long-term conversations, each encompassing **300 turns** and **9K** tokens on avg., over up to 35 sessions.
- Example: Two agents, each initialized with an LLM. Unique **persona statements** are assigned to each agent.
- Create a **temporal event graph** for each agent to mirror real-life experiences
- **Agent architecture** is utilized for each agent, enabling them to effectively memorize and reflect conversation history in ongoing dialogues.



Agent architecture

Source: <https://snap-research.github.io/locomo/>

- The agent is equipped with a **memory** and **reflection** module to retrieve relevant history for dialog generation
- Persona statement and a timeline of casually connected events in their life.



a database of assertions (observations) about each speaker's life and persona

Temporal Event Graph

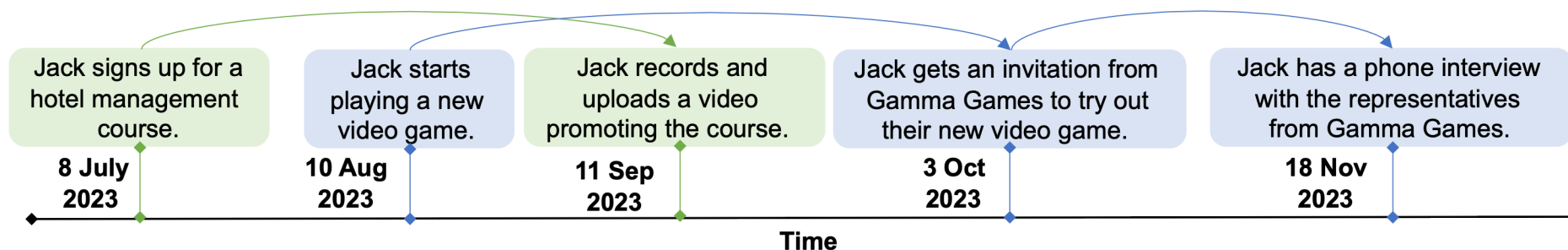
Source: <https://snap-research.github.io/locomo/>

- Each event is generated in accordance with the specified persona and causal connections between events.

Persona Summary

Jack is a 25 year old male who works the graveyard shift at a hotel. He enjoys playing video games in his free time, and listening to pop music. He has a small circle of friends who he meets up with sometimes to have dinner or watch a movie. He also has a few close friends from work. Even though his shift hours are irregular, he still manages to socialize and do activities with his friends. His goal in life is to eventually become a hotel manager.

Causal Events Timeline



Evaluation

Source: <https://snap-research.github.io/locomo/>

- **Question Answering.** Agents need to “recall” past context correctly to integrate relevant information into future responses.
 - Single-hop, multi-hop, common sense & world know., adversarial, temporal
- **Event Graph Summarization.** Agents also need to recognize long-range causal and temporal connections in the dialogues to generate empathetic and relevant responses.
- **Multi-modal Dialog Generation.** Conversational agents need to utilize relevant context recalled from past conversations to generate responses that are consistent with the ongoing narrative.

Findings

Source: <https://snap-research.github.io/locomo/>

- **Long-context LLMs** and **RAG** demonstrate effectiveness in **QA** tasks: improving “memory” capabilities of LLMs, but still significantly lag behind human levels (by 56%), especially in temporal reasoning, (by 73%);

Category	Model	Context Length	Answer Prediction (F1)					Overall
			Single Hop	Multi Hop	Temporal	Open Domain	Adversarial	
Human	Human	-	95.1	85.8	92.6	75.4	89.4	87.9
Base	Mistral-Instruct-7B	8K	10.2	12.8	16.1	19.5	17.0	13.9
	Llama-2-Chat-70B	4,096	19.7	14.4	13.3	15.9	22.1	17.9
	GPT-3.5-turbo	4,096	29.9	23.3	17.5	29.5	12.8	22.4
	GPT-4-turbo	4,096	23.4	23.4	10.4	24.6	70.2	32.1
Long context	GPT-3.5-turbo-16K	4K	31.7	25.4	16.8	27.6	13.1	24.1
		8K	38.8	31.2	21.0	35.0	8.4	25.2
		12K	51.1	40.4	25.0	36.5	6.4	33.5
		16K	56.4	42.0	20.3	37.2	2.1	37.8

Table 2: **Question answering performance** of Base and Long-context models. Optimal performance is in **bold**. Results are based on F1-score for answer prediction; higher is better.

Findings

Source: <https://snap-research.github.io/locomo/>

- **Long-context LLMs has significant hallucinations:** difficulty with *adversarial* questions in QA and event graph summarization task.

(1) Question Answering Task

Based on the given context, write a short answer for the question.

Single-Hop Reasoning

Last night .. we celebrated my daughter's birthday.

Q: Whose birthday did X celebrate?

A: daughter

Commonsense & World Know.

I'm a fan of both classical like Bach and Mozart, as well as modern music like Ed Sheeran's "Perfect".

Q: Would X likely enjoy "The Four Seasons" by Vivaldi?

A: Yes; she likes classical music.

Multi-Hop Reasoning

I visited New York two years ago.

A picture from one of my older travels 

Q: Which places has X visited?

A: New York, Horseshoe Canyon

Adversarial

Sep 5 2020: I'm learning the piano.

Q: When did X start learning violin?
(A) Sep 5, 2020 (B) Not answerable

A: (B) Not answerable

Temporal Reasoning

July 10, 2022: ... I actually started on a book recently since my movie did well!

Oct 6, 2022: ... I finished up my writing for my book last week..

Q: How long did it take for X to finish writing the book?

A: three months

(2) Event Summarization Task

21 January, 2022

I won my first video game tournament last week - so exciting!

The game was called Counter - Strike: Global Offensive ..

23 January, 2022

I start to hang out with some people outside of my circle at the tournament.

24 March, 2022

I'm currently participating in the video game tournament again and it's INTENSE!

Summarize the significant events that have occurred in X's life.

21 Jan: X wins his first video game tournament playing Counter -Strike: Global Offensive with a team.

23 Jan: X starts hanging out with new people he met from outside his circle at the Counter Strike tournament.

24 Mar: X participates in another Counter Strike tournament.

(3) Multimodal Dialog Generation Task

.. Trying out different flavors like chocolate, raspberry, and coconut has been a blast!

Sounds delicious! Are you only trying dairy -free options?

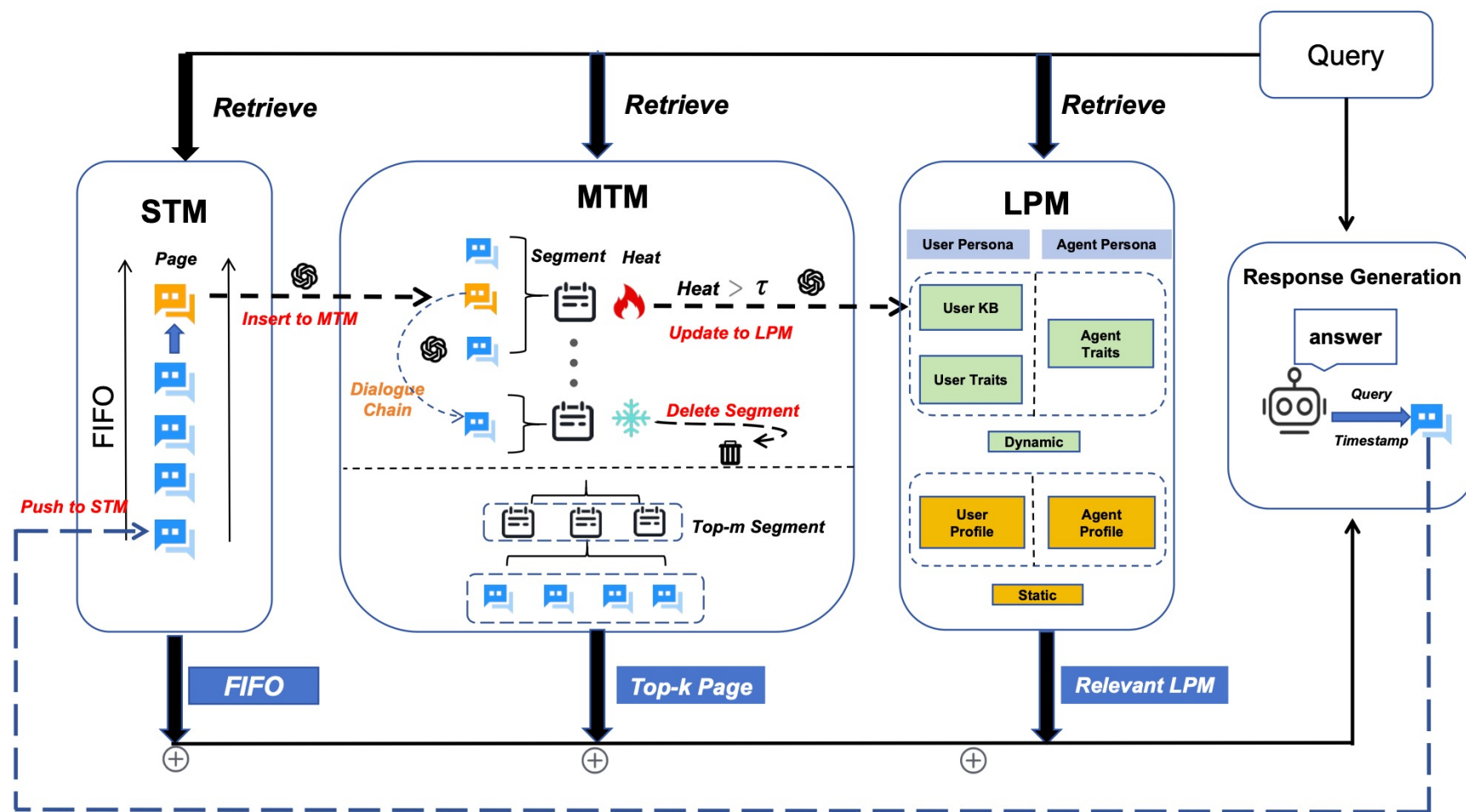
Please generate conversation with appropriate image.

Yeah.. made these dairy-free chocolate coconut cupcakes... 

Related Work

Source: <https://aclanthology.org/2025.acl-long.1126.pdf>

- Memory OS (2025): a hierarchical storage architecture that comprises three levels of storage units:
- Short-term memory,
- Mid-term memory,
- Long-term personal memory.

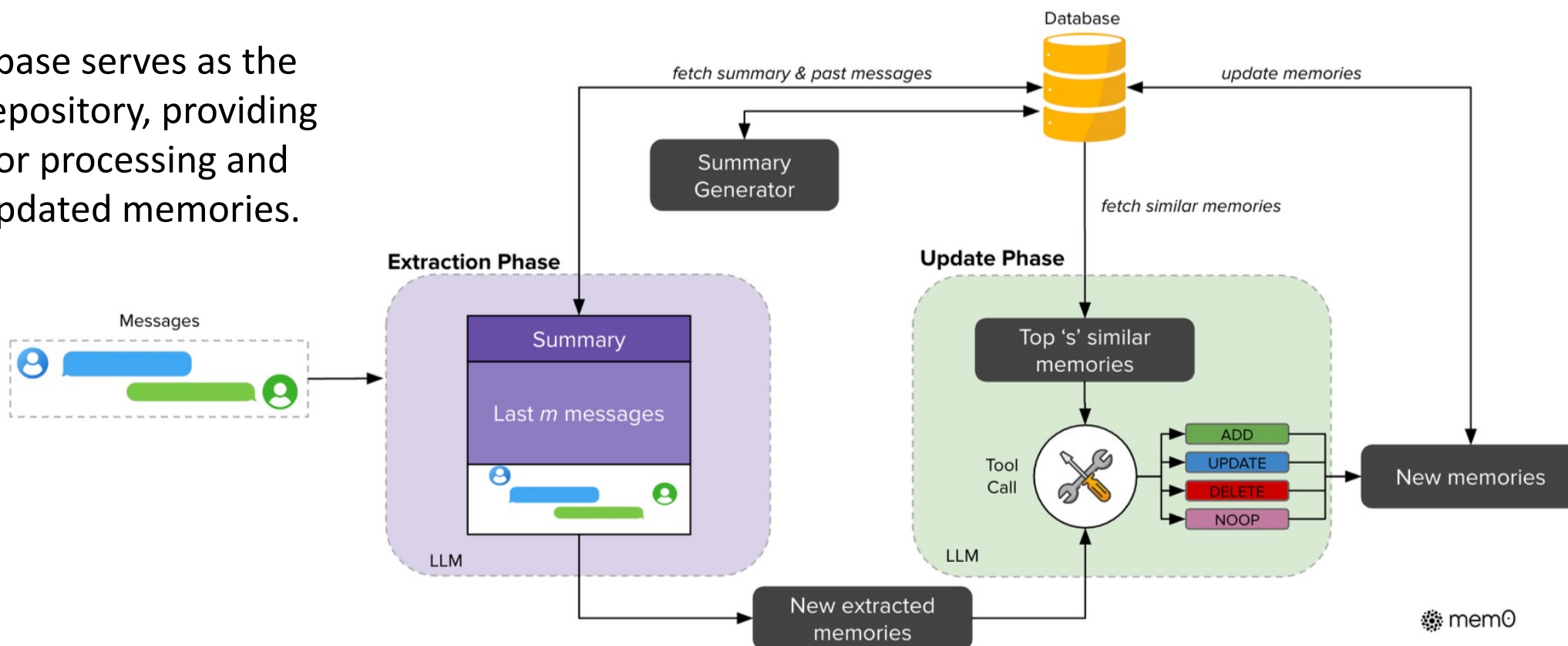


Related Work

Source: <https://mem0.ai/research>

- Mem0: a scalable memory-centric architecture by dynamically extracting, consolidating, and retrieving salient information from ongoing conversations.

The database serves as the central repository, providing context for processing and storing updated memories.



Recap

- Memory System 记忆系统
- Short-Term Memory 短时记忆
- Working Memory 工作记忆
- Memory Mechanism in AI/LLMs 大语言模型中的记忆机制